Relational Information Accounting

Ethics as an accounting frame for whole-system consequences and reachable futures

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Abstract

Actions change the reachable futures and enabling conditions of the systems they affect. This paper derives a calculable whole-system account of those informational consequences. Ethics becomes an accounting frame for consequences that narrower descriptions of an action leave outside their account. An action removes or introduces reachable capabilities, changes the resources required for continuation, and imposes effects along its course. These consequences occur in affected relations whether anyone recognises them or assigns them value. Measurement describes them; decision weights determine how a system responds to them.

Starting from relational existence, admissible transformation, consequential distinction, and grounded expenditure, the paper derives separate loss, gain, resource, and path accounts. No additional biological, sentience, preference, or observer-recognition requirement determines eligibility. The construction retains affected identities through termination and does not assume a common unit across agents. It provides finite calculation functions, proves independence from passive evaluative observation and from decision weights, and recovers a channel-capacity expression for deterministic controllable futures. Worked cases include protective restriction, restoration, and the contrast between disposable game encounters and persistent companions. The account also includes consequences borne by actors and witnesses, providing a structural interpretation of moral injury through damaged self-relations, trust, and continuing participation. It also quantifies a conditional implied decision conversion and the separate consequences of making that conversion explicit.

A further hypothesis interprets human and animal ethical behaviour as compressed expressions of this informational principle. Exact and approximate conditions for preserving decisions under compression are given without claiming to recover a neural implementation. The contribution is an implementable consequence account and its derivation, situated against information ethics, empowerment, and reachability-based impact measures; it does not impose a universal ordering of the resulting decisions.

1 Consequences before decisions

A protective intervention can restrict one person's freedom of action while restoring another's. Both effects occur. Whether the intervention is justified depends on how those effects are weighed. A theory that records only the chosen justification loses the restriction; a theory that records only the restriction loses the protection. A whole-system account must hold both, together with consequences for the actor, witnesses, and the relations that enable later coordination. Ethics brings otherwise omitted consequences into view and studies how agents respond to them through their relative priorities.

The foundational proposition of this paper is:

> An action changes the reachable futures and enabling conditions of affected systems; those changes are accountable across the relations in which they occur.

Here *reachable futures* includes the conditions, resources, and continuing relations through which a future is available. Destroying a capability, making its exercise more costly, and temporarily closing it are different changes. They require different entries even when the final state looks the same.

The resulting lens separates three things. A consequence *occurs* through an interaction. A measurement *represents* that consequence under explicit comparison conditions. A decision *evaluates* the account using weights, priorities, or constraints. An unobserved loss remains a loss. Assigning it zero decision weight does not restore what was removed.

Layer	Determining relation	Role of evaluation
Occurrence	Actual transformation of capabilities and resource conditions	No recognition required
Measurement	Baseline, boundary, identity, horizon, distinction and unit	Specifies the account being taken
Decision	Rule applied to the measured or estimated account	Supplies relative weights and priorities

The proposal locates informational cost in the first layer. The second supplies functions for calculating it. The third allows differences in ethical decisions without changing the consequences already accounted for. An actor's preferences can causally change what the actor does; once the compared actions are fixed, changing an evaluator's preferences changes their ranking rather than their consequence records.

The foundational object is whole-system accounting. The proposed reason for an ethical frame is an accounting deficit: a narrower description holds the actor's objective, immediate output, or direct expenditure while leaving other consequences outside its account. The omitted effects appear as *extra* ethical considerations because the narrower frame has not included their bearers, relations, or continuing costs.

Under the relational ontology, those effects are already changes in the system. A recipient's lost future, an actor's damaged self-relations, a witness's altered trust, and an institution's impaired coordination all have a place in the same informational machinery. They require no separate ethical substance or an observer's assignment of value to become consequences. Ethics therefore becomes an *accounting frame*: it directs attention to effects and incidence that an incomplete operational account leaves out. Relative weights determine how an agent responds to the resulting account.

This is the paper's structural explanation of the need to study ethics. Ethical language and practice can make hidden costs legible and compress sensitivity to them. Studying those practices exposes which consequences a system tracks, which it omits, and how its priorities govern a response. The claim concerns the function of the ethical frame in this ontology; a chronology of the emergence of moral language is a separate empirical question.

Here *whole-system* means seeking causal coverage across affected relations, including the acting system and its internal and social organisation. A finite account still declares its boundary and unresolved effects. The phrase does not certify that every remote consequence has been discovered, and it does not license a single sum over incompatible units or overlapping bearers.

An ethical justification cannot make an incurred cost disappear. An action can be chosen because it protects others while still damaging its actor's trust, self-relations, or continuing participation. Moral injury is therefore a substantive case for the framework: costs can arise within the system that acts, in a witness, and in the institutions that connect them. The relevant

injury is a change of organisation and continuation, beyond an external judgement that an action was good or bad.

Three contributions follow. First, the paper derives an attributed cost account without an observer conferring value on its bearers, including actors and witnesses as affected systems. Second, it supplies actual finite functions for computing the account. Third, it proposes that ethical behaviour can express the underlying principle through compression, without conscious ethical reasoning or literal execution of these functions. The first two stand independently of recovery of a biological mechanism.

2 What the informational foundations supply

The derivation needs structures that can be affected, transformations that can affect them, and distinctions that specify what changes. The preceding papers supply these objects. This section states the dependencies so that an apparent derivation cannot conceal a moral premise in its inputs.

2.1 Existence, representation, and affected frames

Paper 00 treats existence as standing in relation (Parslow, 2026a). A biological organism, a computation, a simulation, and an imagined structure differ in the relations they instantiate and the changes they can sustain. The framework gives them no rank in degree of reality. Being a representation describes a relation between structures; it does not make the representing structure unreal.

Consequences nevertheless belong to the relations actually instantiated. An imagined injury instantiates changes in the imagining system and its representations. It does not thereby instantiate every bodily consequence of the injury represented. A simulated agent can possess persistent capabilities and relationships within its environment; a painted depiction can lack them. Their accounts differ because their continuation structures differ.

An *affected frame* is a structure whose capabilities, expenditure, or continuing relations change under the comparison. "Agent" is used broadly for such a continuing organisation, without requiring a human body, linguistic self-report, or deliberative preference. Where agency or individuality is uncertain, record the candidate partition and its alternatives. Temporal information provides approaches to individuality across scales (Krakauer et al., 2020); it does not supply a moral rank.

The affected frame is also distinct from an observer taking an account. A tree, an unmonitored program, and a collective can each be affected without a further system recognising that effect. If a dimension is instantiated in the affected relation, an observer is unnecessary for its occurrence. If the dimension is absent, a label applied by an observer cannot create it.

2.2 Transformation, capability, and expenditure

Graphic Equalisation supplies admissible transitions and grounding-dependent costs (Parslow, 2026b). Paper 01's well-formedness requirement (Definition 7.13) requires quantities to have a frame and comparison conditions. Its expenditure account (Proposition 7.12) concerns changes and their implementation; it does not identify every expenditure with a decrease of capability.

Seldon Manifolds supplies sets of reachable continuations (Parslow, 2026c). A goal-filtered acceptable manifold (Paper 02, Definition 11.1) is a further construction. The present cost account uses reachable continuations before filtering them by a preferred goal. A frame with no represented preference can still lose a continuation.

Narrativium supplies a measure of consequentially reachable deformation (Parslow, 2026d) (Paper 04, Definitions 2.1 and 3.1). Its distinction between measurement and expenditure is

retained. Its restriction on sums across frames (Proposition 8.1) prevents adding unlike quantities merely because both have been called costs.

These are inherited informational dependencies. The construction below adds accounting definitions and a particular finite realisation. It does not infer an ethical prohibition from a positive expenditure, nor derive all the underlying relational machinery from Shannon's communication theory alone.

2.3 The derivation boundary

The argument has the following dependency order:

1. An instantiated relational organisation has admissible continuations.
2. An action changes the transition relation or the state from which those continuations are available, including the resources required to obtain them.
3. Comparing action and baseline identifies removed and introduced continuations, and changed expenditure, for each affected relation.
4. These differences constitute the informational consequence account.
5. Ethics is an accounting frame that brings omitted effects and their bearers into the comparison. Agents respond through represented information, compressed dispositions, and relative priorities. A decision rule evaluates the account; its judgement does not constitute the recorded changes.

Steps 1–4 supply the whole-system construction. Step 5 locates ethics as an accounting frame within that construction and introduces the hypothesis connecting behaviour with the account. Nothing in this chain requires a morally approved species, a sentience threshold, a represented preference, or an observer's award of value. Grounding an actual quantity is still necessary; an unknown quantity is not a known zero.

3 The comparison contract

Cost relative to an action requires a specified alternative. Removing a bridge has different consequences compared with leaving it open, replacing it, or closing it before a collapse. A baseline answers which comparison is being made. It does not decree which comparison must govern every decision.

Definition 3.1 (Comparison contract)**.** *A contract θ specifies:*

1. *the compared action a and baseline b, including subsequent policies;*
2. *the causal boundary and affected identities $\mathcal{I}_\theta$, with continuity conditions across both branches;*
3. *a horizon and observation times or intervals;*
4. *each frame's consequential outcome space Ω_i, an aligned outcome map, and a nonnegative measure μ_i where defined;*
5. *resource dimensions, units, payers and attribution rules;*
6. *model uncertainty and conditions under which the account is complete.*

A contract represents the relations being compared. It does not bring their changes into existence. Moving a measurement boundary changes what an account contains; it does not undo a consequence outside that boundary. Claims of completeness therefore require a causal argument, rather than merely a short list of preferred recipients.

3.1 Consequential distinctions

Two descriptions count as the same outcome when the declared grounding identifies them with the same relevant continuing capability. Renaming an option cannot create an additional option. Conversely, two equally numerous outcome sets need not offer the same capabilities. A chance to leave, a chance to work, and a chance to obtain food are not interchangeable because each set has three members.

The outcome quotient specifies the resolution of an account. Its justification must be inspectable in the affected dynamics. A numerical outcome mass can express a grounded amount of an available capability or a sampling convention. A mass chosen because a decision-maker likes one outcome more than another is an evaluation weight and belongs in the decision rule. Calling it μ_i would hide valuation inside the calculation.

A measure can therefore be objective within its declared grounding without being the unique measure of every consequence. The full set changes retain identities that a scalar summary loses. A coarse account that merges a relevant distinction is incomplete at that finer resolution; its zero does not establish that no cost occurred.

3.2 Identity, time, and attribution

Use the union of affected identities across branches. The loss of a frame cannot remove that frame from the population being compared. A created successor is matched to the baseline frame only where the continuity conditions establish that match. Similar appearance, a copied name, or a new numerical identifier is insufficient by itself.

For shared resources, separate physical expenditure from its attribution. A single shared bill can be represented once with links to all affected frames. Allocating that bill among payers requires either grounded incidence or an explicit allocation convention. Summing the full bill once for every claimant manufactures expenditure. Collective and member accounts can coexist, but their entries do not become disjoint by being listed separately.

The affected population includes the actor whenever its action changes its own organisation or continuations. It also includes witnesses, dependants, and institutions where the action changes their enabling relations. Direct bodily impact does not exhaust causal incidence. A memory, expectation, permission, or trust relation can change later admissibility and expenditure, and therefore belongs in the model where that effect is grounded. A represented judgement inside an affected system is a causal state; it is different from the passive external evaluation excluded in Proposition 5.1.

Horizon and path are likewise distinct. A restored system can have the same terminal capabilities while having undergone an interval of restriction. The comparison must state whether it takes a terminal account, an account along a course, or both. No undisclosed preference for the present over the future is introduced by the time index.

4 Deriving the cost functions

The account starts with what remains possible. Expenditure and time enter as additional typed dimensions because neither is recoverable from a final option count alone.

4.1 Reachable futures and their changes

Let S_x be the state space for branch $x \in \{a, b\}$ and let T_x be its admissible transition relation. Let s_x be the post-intervention starting state. For frame i, the map π_i assigns aligned consequential outcomes to states; $\perp$ means that the state instantiates no continuing outcome for that frame.

For a fixed-step horizon h define

$$\mathcal{R}_i(x;h) = \{\pi_i(s) : s_x \xrightarrow[T_x]{h} s,\ \pi_i(s) \neq \bot\}. \tag{1}$$

A horizon of *up to* h steps uses the union over $0 \leq t \leq h$ instead. These answer different questions. Terminal persistence must be encoded if an exact-step account is intended to include terminated world states.

Resources that constrain feasibility belong in the state or admissibility relation: for example (s, β, t) can carry remaining budget β and time t. Otherwise a traversal can count an option that its bearer cannot afford. An institution's permission and a system's physical capacity can likewise be separate state components when either changes the actual reachable set.

Definition 4.1 (Lost and gained futures)**.** *For aligned outcome spaces under one contract, define*

$$\mathrm{Lost}_i(a \mid b) = \mathcal{R}_i(b) \setminus \mathcal{R}_i(a), \qquad L_i(a \mid b) = \mu_i(\mathrm{Lost}_i(a \mid b)), \tag{2}$$

$$\mathrm{Gained}_i(a \mid b) = \mathcal{R}_i(a) \setminus \mathcal{R}_i(b), \qquad G_i(a \mid b) = \mu_i(\mathrm{Gained}_i(a \mid b)). \tag{3}$$

If μ_i is unavailable, retain the sets and mark their numerical measures undefined. For finite distinct outcomes, counting measure is a calculable specialisation.

The functions follow directly from the two reachable sets. They require no preference ordering on their members. The pair (L_i, G_i) is retained rather than cancelled: an action can close an exit and open an unrelated opportunity. When finite measures exist,

$$\mu_i(\mathcal{R}_i(a)) - \mu_i(\mathcal{R}_i(b)) = G_i - L_i. \tag{4}$$

The difference is a net capability summary. It omits which futures were replaced.

4.2 The information-theoretic quantity

Counting consequentially distinct futures connects the construction to information theory. In a deterministic control channel, let U denote legal inputs and $Y = f(U)$ their consequential outputs. Empowerment is the channel capacity

$$E = \max_{p(U)} I(U;Y), \tag{5}$$

introduced as an agent-centred measure of potential control by Klyubin et al. (2005). Its units are bits when logarithms have base two.

Proposition 4.2 (Finite deterministic capacity)**.** *If a deterministic channel has $m \geq 1$ distinct outputs and each is attainable by a legal input, then $E = \log_2 m$.*

Proof. Determinism gives $H(Y \mid U) = 0$, so $I(U;Y) = H(Y) \leq \log_2 m$. Choose one input for each output and assign these m inputs equal probability. The outputs are then uniform and the upper bound is attained. □

Thus a nested restriction from m_b to $m_a \geq 1$ controllable outcomes removes

$$\Delta E = \log_2 m_b - \log_2 m_a = \log_2(m_b/m_a) \tag{6}$$

bits of attainable distinction. This recovers a known capacity expression; it makes the informational content of the finite loss concrete. For non-nested sets, capacity difference alone conceals replacements. For stochastic systems, possible outcomes and controllable outcomes differ: an uncontrollable lottery can have many reachable results without high channel capacity. Use the actual channel when measuring empowerment, and reachability when measuring possibility.

No channel with zero attainable outputs is assigned $\log_2 0$. Extinction is recorded through lost continuations and identity, while its capacity entry is undefined. A deterministic one-output channel has zero bits of control; removing its sole continuing outcome can still produce a nonzero set loss. Capacity is one dimension of the account, rather than a complete measure of existence.

Shannon uncertainty alone is also insufficient. A string of random bits does not acquire ethical priority through its entropy. Relevant information here is the organisation of consequential continuations and their enabling relations. Communication theory supplies the capacity calculation (Shannon, 1948); the relational grounding supplies what the channel connects and whom its alteration affects.

4.3 Resource burdens and access prices

Let $q_{ir}(x)$ be branch x's actual or predicted expenditure borne by frame i in grounded resource dimension r. The incremental burden is

$$K_{ir}(a \mid b) = q_{ir}(a) - q_{ir}(b). \tag{7}$$

This is signed: a negative entry records saved expenditure. Energy, time, attention, computational work, and access payments require their own instantiated units and incidence. A grounding that supports joules does not automatically support a quantitative measure of attention.

An *access price* is different. If $p_{ir}(y \mid x)$ is the least expenditure required for frame i to attain outcome y from branch x, a change in that price can burden the frame before it attempts the outcome. Retain the price profile over aligned outcomes where it is defined. A feasible set can remain unchanged while all its members become more expensive. Expenditure in Eq. (7) is what was paid; an access-price change is what would have to be paid. The two must not be silently substituted.

The substrate does not imply a universal exchange rate between these resource entries and lost capabilities. Its information-erasure discussion also does not license charging a fixed Landauer cost for every logical or relational change. Logical irreversibility and physical implementation conditions must be supplied before such a bound is used.

4.4 Effects along the course and restoration

For aligned comparison times t, calculate

$$L_i(t; a \mid b) = \mu_i\big(\mathcal{R}_i(b; t) \setminus \mathcal{R}_i(a; t)\big). \tag{8}$$

Here $\mathcal{R}_i(x; t)$ describes capabilities from the state reached at t, under the same declared continuation horizon. These sets are conditional on a realised or specified course; uncertain courses require a profile or distribution of accounts. They are not automatically the unconditional terminal set in Eq. (1).

The matching gain profile is

$$G_i(t; a \mid b) = \mu_i\big(\mathcal{R}_i(a; t) \setminus \mathcal{R}_i(b; t)\big). \tag{9}$$

A capability made available only temporarily can disappear from a terminal snapshot while remaining a gain along the course. Retain both profiles without cancelling unrelated losses and gains.

Retain loss and gain profiles as the primary path account. Where a clock and an integration measure ν are grounded, define exposure

$$D_i(a \mid b) = \int L_i(t; a \mid b)\, d\nu(t). \tag{10}$$

For piecewise constant intervals of duration Δt_k, this is $\sum_k \Delta t_k L_i(t_k; a \mid b)$. Its units are capability units multiplied by clock units. It is not another terminal loss in the same units. Similarly, where the same clock is grounded, gain exposure is $D_i^+(a \mid b) = \int G_i(t; a \mid b)\, d\nu(t)$. It has the corresponding capability-time units and remains separate from loss exposure.

Exact restoration at T can give $L_i(T) = 0$ while $D_i > 0$ and expenditure is positive. Paper 02b's deprivation-and-restoration example (Remark 2.1) motivates this distinction. An imposed arousal variable is not sufficient evidence of experienced suffering; the structural point is that a terminal snapshot can omit an intervening restriction.

This historical account does not posit an invisible historical force acting on a restored state. It compares the courses that occurred. It also does not require that a recipient retain a memory of the restriction. An external record makes the account recoverable; it does not constitute the past transformation.

4.5 Repair feasibility

Define a restoration condition B_i on continuing capabilities and relations. Let $\mathcal{P}_i$ be the admissible repair paths from the post-action state to a state satisfying B_i. For one grounded resource unit, define

$$\rho_i(a \mid b) = \inf_{p \in \mathcal{P}_i} q_i(p), \qquad \inf \varnothing = +\infty. \tag{11}$$

A finite nonnegative-cost graph permits a minimum-cost path calculation. With multiple incompatible units, retain the feasible resource vectors or their Pareto frontier; a scalar shortest path requires a declared conversion.

A repair estimate is a prospective liability. It is not expenditure already incurred, and restoring a terminal capability does not reverse prior exposure. If repair is carried out, its actual expenditure belongs to the repaired branch's resource record. Adding the estimate again would charge the same repair twice.

4.6 The complete account and its finite realisation

Definition 4.3 (Informational consequence account)**.** *For a contract θ, the account is the typed family*

$$\mathcal{C}_\theta(a \mid b) = \left\{ \begin{array}{c} (\mathrm{Lost}_i, \mathrm{Gained}_i, L_i, G_i, (K_{ir})_r, (L_i(t))_t, (G_i(t))_t, \\ (p_{ir}(\cdot \mid a), p_{ir}(\cdot \mid b))_r, \rho_i) \end{array} : i \in \mathcal{I}_\theta \right\}, \tag{12}$$

together with identity, shared incidence, grounded stochastic-transition or control-channel records where applicable, and uncertainty records. Support sets do not determine outcome probabilities or control reliability: changes in those records require their own aligned comparison and are not represented by a zero set difference. A component is included only where its dimension is instantiated; numerical absence is marked undefined. Informational cost denotes the losses and burdens in this account; gains and savings are retained alongside them. An ethical accounting frame selects and examines consequences and incidence; decision rules evaluate the account without redefining the occurrence of its entries. A frame that leaves a consequence out does not make that consequence absent from the system.

This is a family of actual functions, not a single untyped score. Appendix A prints a finite Python realisation. Its graph traversal computes Eq. (1); outcome comparison computes Eqs. (2)–(3); resource comparison computes Eq. (7); interval accounting computes Eq. (10) and the matching gain exposure; and nonnegative shortest-path search computes Eq. (11). A final function evaluates explicitly supplied features and weights after the consequence calculation.

The implementation accepts measured graphs and outcome maps. It does not pretend to infer a complete causal world from raw observations. Frame identity, remaining budget, and clock conditions belong in the supplied state model when they affect admissibility. A missing

frame key means a *known* empty continuation set; a missing measured quantity must not be passed as an empty set. The same restriction applies to resource entries omitted as known zero.

For finite graphs, stepwise reachability visits at most all supplied edges at each of h steps, giving an $O((h+1)(|V|+|E|))$ bound including materialisation, before outcome processing. Hash-based set comparison is linear in the represented outcome sets in the usual expected case. Repair search uses Dijkstra's algorithm with nonnegative scalar edge costs. Augmenting states with budgets or enumerating long courses can greatly increase the graph: calculable does not mean cheap at every scale.

Finite adjacency iterables are materialised so that repeated visits preserve the supplied transition relation. The numerical implementation uses finite arithmetic: out-of-range computed masses, exposures, scores, or explored repair-path costs raise an error. They are not reported as measured infinite burdens or as evidence that repair is impossible.

The functions realise the finite support, resource, path, and repair specialisation. Access-price profiles use repair search separately for each target outcome; path accounts across uncertain courses require those courses to be supplied. A stochastic channel requires channel-capacity machinery beyond the deterministic function. These extensions change the measurement implementation, without introducing a moral eligibility condition.

5 What follows from the construction

The construction earns its main claims under explicit conditions. In particular, observer independence concerns whether the consequence occurs; it is stronger than leaving a weight variable out of a formula.

Proposition 5.1 (Independence from passive evaluation)**.** *Fix the actual affected transition relations, including any recorded stochastic laws or control channels, starting states, expenditures, and continuity conditions. Add, remove, or change an evaluator whose observation and judgement do not causally alter those relations. The removed and introduced continuations and incurred expenditures are unchanged. Where the measurement contract is also held fixed, $\mathcal{C}_\theta(a \mid b)$ is unchanged.*

Proof. Occurrence is determined by the affected starting states and admissible transitions. By hypothesis these inputs and the expenditure records do not change with evaluation. Hence the sets of admissible paths do not change, their aligned outcome images do not change, and their set differences do not change. The resource differences and course profiles likewise retain their inputs. No report or judgement is an argument of these operations. □

A closed exit remains closed without a witness. That is the causal content of the proposition. An evaluator who opens the exit, imposes a restriction, or pays for an option is an additional actor; the causal comparison has changed. An evaluator who uses a coarser outcome map changes their account's resolution. Neither case makes recognition the source of the earlier cost.

Proposition 5.2 (Independence from decision weights)**.** *For fixed θ, a and b, replacing an evaluation rule V_w by $V_{w'}$ does not change $\mathcal{C}_\theta(a \mid b)$.*

Proof. The account uses transition, outcome, expenditure and comparison inputs. Decision weights occur only in the subsequent evaluation map $V_w : \mathcal{C}_\theta \mapsto$ decision score or ordering. Replacing that map leaves its argument fixed. □

Proposition 5.3 (Representation invariance)**.** *Suppose two groundings are related by bijections preserving affected identity, starting states, admissible transitions, consequential outcome identity, resource incidence, units and numerical expenditures, time alignment, outcome measures, access-price profiles, and any recorded stochastic laws or control channels. For the repair component,*

also preserve admissible repair transitions, their numerical edge costs, and restoration predicates. Their consequence accounts agree under those bijections.

Proof. A transition-preserving bijection carries each admissible path to an admissible path of the same declared length. Its aligned outcome bijection therefore carries reachable sets to reachable sets and commutes with set difference. Measure preservation gives equal losses and gains. Resource and time preservation give equal burdens and profiles. Repair paths correspond with equal expenditure and restoration conditions, so their infima agree. □

This states what substrate independence buys: the procedure and its results survive changes of representation that preserve the relevant structure. Two embodiments with different resource demands or continuations need not have equal accounts merely because both are called agents. A simulation label cannot by itself discount an isomorphic consequence.

Proposition 5.4 (Termination remains in the account)**.** *If an affected identity has a nonempty baseline reachable set and no continuing action outcomes, its lost set equals the baseline set. Its loss remains in the account even if the post-action population omits that identity.*

Proof. The comparison population is the union across branches. For that identity, $\mathcal{R}_i(a) = \varnothing$, so $\mathrm{Lost}_i = \mathcal{R}_i(b) \setminus \varnothing = \mathcal{R}_i(b)$. The account is indexed by the comparison population rather than by the surviving census. □

A copy contributes a continuation of the original only where the continuity conditions support it. The framework thus makes replacement a question about actual retained relations. It cannot erase a loss through renaming its bearer.

6 Decisions and relative weightings

A derived cost does not force one action in every conflict. Protecting a recipient can require restricting an attacker; preserving a system can require spending resources that would otherwise remain available elsewhere. The account exposes what each choice changes. A decision rule makes its priorities inspectable.

For numerical components in a declared evaluation space, one example is

$$V_w(a \mid b) = \sum_i \alpha_i L_i - \sum_i \beta_i G_i + \sum_{i,r} \gamma_{ir} K_{ir} + \sum_i \delta_i D_i - \sum_i \eta_i D_i^{+}, \tag{13}$$

with an action selected by minimising V_w among feasible alternatives. Weights express relative priorities and any chosen conversions. When an outcome identity matters independently of its count, replace the corresponding scalar by an outcome-level feature vector. Lexicographic rules and constraints can also evaluate the same account. A hard constraint is a decision condition; it is not evidence that the excluded action had no measurable consequences.

Resources, capability counts and capability-time exposure cannot be added as physical quantities without conversion. Equation (13) is a declared evaluation function. A conversion grounded in a common physical quantity differs from a stipulated priority between unlike quantities, as Paper 04 explains (Remark 8.3). Both can be used if their status is stated. Neither silently changes the underlying quantities.

Proposition 6.1 (Conditional dominance)**.** *In a common declared feature space, let losses, burdens and exposure have positive signs, and gains negative signs. If $z_j(a) \le z_j(a')$ for every feature j, then $w \cdot z(a) \le w \cdot z(a')$ for every nonnegative w. The inequality is strict if a strictly smaller feature receives positive weight.*

Proof. Each term $w_j(z_j(a') - z_j(a))$ is nonnegative. Their sum has the stated sign, and is positive under the strictness condition. □

This is agreement within a class of evaluators, not a proof that every possible agent must use nonnegative weights. For a declared admissible weight set W, robust preference for a over a' means $\sup_{w \in W} w \cdot (z(a) - z(a')) \leq 0$. If the inequality fails, the disagreement can concern weights even when every evaluator accepts the measured account.

Consent has a place in both layers. It can causally change coordination, permission, enforcement cost, and actual reachability. It can also receive weight in a decision. It does not erase a remaining restriction by changing its name. Similarly, inability to protest does not remove a bearer from the account. Paper 01 already distinguishes occurrence of harm from its recognition (Remark 10.2); this construction makes the distinction explicit in the calculation.

7 Worked cases

7.1 Protection with a calculable restriction

Let frame A have six baseline outcomes $\{1, 2, 3, 4, 5, 6\}$ and frame B one baseline outcome $\{x\}$. A restriction leaves A with $\{1, 2, 3\}$ and gives B the outcomes $\{x, y, z, u\}$. Under counting measure,

$$L_A = 3, \quad G_A = 0, \quad L_B = 0, \quad G_B = 3.$$

Suppose the action also imposes one resource unit on B, while the baseline imposes none. Then $K_{B,r} = 1$. A declared decision score is

$$V_w = 3\alpha_A + \gamma_{B,r} - 3\beta_B. \tag{14}$$

For $(\alpha_A, \gamma_{B,r}, \beta_B) = (1, 1, 2)$ the score is -2; for $(2, 1, 1)$ it is 4. The baseline has score zero in both accounts. The two evaluators choose differently while agreeing on every measured consequence.

The appendix computes these values by set comparison, rather than defining loss as a preference. For deterministic controllable channels, A loses one bit of capacity, $\log_2(6/3) = 1$, while B gains two bits, $\log_2(4/1) = 2$. Counts and bits are alternative summaries with explicit units; neither is added to the other as though it were another independent loss.

Equal counts can conceal a different case. Replacing $\{1, 2, 3\}$ with $\{4, 5, 6\}$ leaves capacity and cardinality unchanged while recording three lost and three gained outcomes. A rule that uses only net count ignores that substitution. The fuller account makes the omission visible.

7.2 Short-term options and continuing viability

An action can open many immediate options by exhausting a budget required for later continuation. A short horizon registers expansion; a longer horizon registers contraction. This is a change of comparison question, not an inconsistency in the set operations. Include remaining resources in the state and publish the horizon profile when short and long accounts differ.

The same applies to delegation and automation. A recipient can gain access to an outcome while becoming dependent on a provider whose withdrawal closes it. The continuation graph contains that dependency. Counting only the currently available menu omits the provider relation and the cost of maintaining access.

7.3 Restoration without cancellation

Suppose the baseline maintains four capabilities through three unit-length intervals. The action permits only one through the first two intervals and restores all four in the third. The loss profile is $(3, 3, 0)$, terminal loss is zero, and exposure is $D = 6$ capability-time units. If the restoration consumes two resource units, that expenditure is a separate entry.

Conversely, an action can temporarily add one capability to the same baseline through the first two intervals, then return to the original four. Its gain profile is $(1, 1, 0)$ and gain exposure is

$D^+ = 2$ capability-time units, despite zero terminal gain. The function `gain_profile` calculates this matching course account.

The account establishes temporary restriction, exposure and expenditure. Whether these quantities correspond to pain, distress, or another experienced effect requires a grounding for that effect. A conscious preference to remember the restriction is unnecessary for recording the structural loss. Byte-identical restoration does not make the two courses identical.

7.4 Termination and replacement

A frame with baseline continuations $\{r, s, t\}$ is terminated in the action branch. Its continuing outcome set is empty and its loss is three. A newly created frame with three outcomes does not cancel the loss by cardinality. If a repair or transfer preserves the original's identified relations, the continuity contract can establish which outcomes remain available. The account then changes because retained structure has changed, rather than because a new entity was announced as equivalent.

The same rule applies to biological and computational cases. Their expenditures and continuity relations can differ substantially. No category supplies a blanket exemption or a universal equality of numerical costs.

7.5 Disposable encounters and persistent companions

Games offer a particularly clear comparison because their continuing relations can be inspected in the program. A player can treat the loss of one companion as more consequential than the removal of a thousand encounter enemies. The framework distinguishes two mechanisms: the underlying structures can lose different futures, and a player can weight their recorded losses differently. The first mechanism exists before the second.

Consider a finite game with an encounter character e. In both branches the encounter ends and the character has no persistent state, later action, social relation, or continuing world role. Let the comparison concern the state after that boundary. The baseline resolves the encounter without the depicted killing; the action resolves it with the depicted killing. If no continuing outcome of e is instantiated in either branch, then

$$\mathcal{R}_e(b) = \mathcal{R}_e(a) = \varnothing, \qquad L_e = G_e = 0. \tag{15}$$

The zero concerns actual absent continuing futures. It is not a low weight assigned to a rich persisting agent. If the wider world, player state, and resource expenditure also agree, those account entries agree as well.

Now let a companion c have three distinct reachable continuing outcomes: a later joint action, an ongoing exchange, and a persistent world response. Specify $\mathcal{R}_c(b) = \{r_1, r_2, r_3\}$ and $\mathcal{R}_c(a) = \varnothing$ under an outcome map that distinguishes these possible continuations. Then $\mathrm{Lost}_c = \{r_1, r_2, r_3\}$ and $L_c = 3$ under counting measure. This toy count concerns distinct outcomes, rather than counting three concurrent features of one outcome as if they were separate channel outputs. Effects on a player and other characters belong to their corresponding accounts. They need not first judge the death meaningful for those continuations to be removed. Player valuation explains an additional part of the resulting response.

Equation (15) requires its stated boundary. An encounter character can have intermediate options, a restriction interval, or effects on another agent. If killing it early removes those options, its path account can be positive even when its terminal continuing account is zero. If it learns, persists, or sustains relations outside the encounter, its future set is different. "Designed to be killed" is therefore evidence about a role, not a mathematical proof of zero cost. The code and actual continuation relations settle the case.

Likewise, a depiction can carry substantial consequences in a player's relationships, retained game history, or other instantiated structures even if the target has no persistent organisation.

A represented life story can become part of such a structure. The account follows the relations that exist rather than granting or removing consequences through the adjective "fictional."

A thousand characters satisfying Eq. (15) do not create a thousand lost continuing lives merely because the depicted acts share the label killing. In a common game-world account with identical continuing outcomes, the loss remains zero. Positive intermediate or resource entries, where present, remain positive; aggregation requires their common units and incidence. This explains how the informational cost of a disposable encounter can be nearly trivial while a companion's loss is substantial.

The 1990s video-game controversy supplies historical context. The US Senate's activity report documents the 1993 and 1994 hearings on violent game content (United States Senate Committee on Governmental Affairs, 1995). A content label identifies what is depicted. The present theory asks which actual futures, relations, and resources the depicted act removes. The two classifications need not coincide. Research on game-related guilt and moral disengagement addresses players' responses to such framing (Hartmann and Vorderer, 2010); that response is a measurement or valuation phenomenon, not proof of the target's underlying continuation account.

8 Moral injury as a whole-system consequence

Moral injury shows why an account confined to an action's external recipient is incomplete. The apparent extra ethical cost is an omitted internal or relational cost of the system. An actor can alter their own continuing organisation by doing something, being unable to prevent it, or receiving information about it. Witnesses and institutional relations can also bear persistent consequences. The resulting costs need not disappear when the action receives a favourable ethical judgement.

Litz and colleagues describe lasting consequences following events that violate deeply held moral expectations, including perpetration, failure to prevent, and witnessing (Litz et al., 2009). Shay emphasises high-stakes betrayal by legitimate authority and impaired trust (Shay, 2014). These accounts supply the phenomenon to explain. The present framework proposes an informational interpretation of its changes, rather than substituting a capability count for a clinical construct.

8.1 Self-relations and trust are enabling structure

A person's expectations about themselves and others can enable continuations: accepting help, cooperating, making commitments, exercising a role, or relying on an institution. These possibilities depend on actual embodied, cognitive, and social organisation. If an event changes that organisation, it can close continuations or increase the effort needed to sustain them. The account uses the same loss, access-price, expenditure, and path functions already defined. No extra moral-injury eligibility rule is needed to record the change.

Unchanged support sets do not establish unchanged reliability or burden. Where trust changes the probability of successful cooperation while leaving it possible, retain the grounded transition probabilities or control channel alongside the support account. Where an experienced or physiological burden is instantiated, its own grounded dimension is required. The finite set functions alone do not certify the absence of those effects.

This is the proposed mechanism at the structural level. The event changes retained information and enabling relations; those changes alter subsequent admissibility and costs. A self-model or expectation matters because of what it causally enables or restricts. Mere disagreement between two reported beliefs does not establish that a capability was lost. Conversely, a damaged relation can matter even when its bearer cannot articulate an ethical explanation.

Internal moral commitments can enter this causal process without becoming externally imposed accounting axioms. They are part of the organisation of the particular system being studied. A commitment violated by an event can change that system's future behaviour, relations, and expenditure. Another system with different commitments can undergo different changes. The functions remain the same while their grounded inputs differ.

8.2 A chosen action can retain an internal bill

Extend the protective example of Section 7.1 with an actor P. Let its baseline continuing outcomes be $\mathcal{R}_P(b) = \{u, v, w\}$ and its post-action outcomes be $\mathcal{R}_P(a) = \{u\}$, where the outcome map distinguishes grounded continuations involving cooperation and participation. If the action also imposes two additional resource units on P, then $L_P = 2$ and $K_{P,r} = 2$. These are illustrative consequences of the specified model, not numerical findings about a person.

The original protective account remains: $L_A = 3$, $G_B = 3$ and $K_{B,r} = 1$. The expanded decision score includes

$$V_w = 3\alpha_A + \gamma_{B,r} - 3\beta_B + 2\alpha_P + 2\gamma_{P,r}.$$

For weights that place sufficient priority on protection, this action can still be selected. Its selection does not cancel the actor's loss. The full account explains how an action regarded as necessary or justified can carry a persistent cost for the system that performed it. Establishing moral injury in a particular person requires evidence about the event, the affected organisation, and its consequences; the toy account supplies no diagnosis.

The same reasoning covers a witness whose continuing trust is damaged, or a person unable to prevent an event. Attribution asks what actually changed and through which relations. It does not require the affected person to have caused the original external injury. An institution's practical ability to coordinate can likewise change when trust relations are damaged. Individual and collective entries must retain their overlap rather than be indiscriminately added.

8.3 The trolley switch: external consequences and operator state

The trolley problem makes this separation explicit. Thomson's treatment already compares the trolley driver and a bystander at the switch, including their responsibilities and discomfort (Thomson, 1985). The present account uses the established dilemma to calculate separately attributed consequences.

Fix a as operating the switch and b as leaving it unchanged, with the usual stipulated external outcomes: a saves five people and kills one who would survive under b. For each saved person k, retain the grounded gain $G_k(a \mid b)$; for the person killed, retain $L_6(a \mid b)$. The headcount specifies neither the lost continuations nor a universal exchange rate between these persons. Their identified futures, external resource changes, witnesses, and institutional effects belong in the external account under its declared boundary.

Let x_P describe the operator's initial organisation, including commitments, self-model, experience, knowledge, and enabling relations. Write the operator's attributed component as

$$\mathcal{C}_{\theta,P}(a \mid b; x_P). \tag{16}$$

This is calculated with the same functions, using the operator's branch-specific continuations, expenditures, path profiles, and repair conditions. It is not a casualty count converted into guilt. Operating the switch can change one operator's continuing self-relations or impose substantial expenditure, while another operator undergoes different changes under the same external outcome. Declining to act can likewise impose an internal burden; the baseline is not presumed free of consequences.

For a finite illustration, fix the same external branches and give two possible operator states the baseline outcome set $\{u, v, w\}$. In state x_P, switching leaves $\{u\}$ and adds two resource

units, yielding $L_P = 2$ and $K_{P,r} = 2$. In state x'_P, switching leaves $\{u, v, w\}$ with unchanged expenditure, yielding $L_P = 0$ and $K_{P,r} = 0$ in these dimensions. Other dimensions remain to be checked. These are specified toy inputs, not measured psychological values. They demonstrate that operator cost is calculable and state-dependent while the fixed external losses and gains remain the same.

The separation is one of attribution, not causal isolation: what the operator knows or believes about the external event can change their internal transition. Nevertheless, the external impact alone does not determine the operator's cost. Their feeling of justification cannot erase another person's loss, and their internal burden cannot be inferred solely from the number saved or killed. A decision can therefore favour switching while retaining a substantial operator cost. Relative priorities decide how these separate entries enter the choice; no single preferred answer to the dilemma follows from accounting alone.

8.4 Psychopathic traits and differences in ethical aperture

The same separation permits an account of differences associated with psychopathic traits. Aperture retains its inherited meaning: the input gate and its represented dimensions, resolution, and extent (Paper 01, Definition 7.2; Paper 02, Remark 5.3). An *ethical aperture* is the application of that structure to ethically relevant distinctions and their fit to acceptable future shapes (Paper 02, Remark 5.4). It is a shape, rather than one quantity of ethical sensitivity. Different frames can represent different bearers, kinds of consequence, degrees of distinction, and horizons. An ethical cost dimension absent from a frame is undefined there, rather than present with value or weight zero. This is absence to that frame, not absence of the affected relation from the world.

The shape of what remains acceptable adds the flexibility described by this application. Let $\widehat{\mathcal{S}}_j(x)$ be actor j's represented feasible continuations from state x. Where the actor holds an operative ethical acceptability predicate $\Gamma_{E,j}$, define

$$\mathcal{E}_j(x) = \{y \in \widehat{\mathcal{S}}_j(x) : \Gamma_{E,j}(y \mid x)\}. \tag{17}$$

This is a conditional filter on the represented manifold. It is distinct from the input aperture that determines which distinctions can enter that manifold, and from the subsequent weights ordering its members. It is not a moral eligibility gate in the consequence account.

Holding the represented manifold fixed, more permissive ethical conditions can admit more future shapes; readily revised or context-dependent boundaries can make the operative ethics flexible. An actor may instead retain only a few such conditions, or no effective ethical restriction over the compared domain. In that last case the ethical filter excludes nothing there: effectively $\mathcal{E}_j(x) = \widehat{\mathcal{S}}_j(x)$. This does not require an ethical predicate to be explicitly represented when none is operative, nor make physical, resource, or other nonethical constraints disappear. Comparative ethical priorities can also remain weak or strong within the admitted region. Few constraints, low weights, and absent dimensions are different possible structures.

Under the inherited fit/adaptive-resolution hypothesis (Paper 02, Hypothesis 8.1), an already acceptable continuation can attract little further ethical scrutiny: it passes with room at the represented grain. A continuation near a held boundary can instead attract refinement or pressure to reshape the acceptable region. What lies outside a represented dimension or horizon never reaches that fit comparison. These structural possibilities and predictions alone do not establish a globally wider aperture or a generally more flexible person; comparisons must state the dimensions, horizons, and held conditions. The application uses specified trait profiles under matched conditions; psychopathy or sociopathy labels alone do not supply these inputs.

On this hypothesis, the patterns grouped under psychopathy can be described as a characteristic family of ethical-aperture shapes and acceptable-region boundaries. Membership in that family is an empirical question, assessed through represented distinctions and held constraints across situations. The hypothesis does not identify every such profile with one identical shape.

The proposed definition is intrinsic: its defining properties are the person's operative distinctions, acceptability boundaries, and their dynamics within that person's actual relational organisation. They exist whether or not an observer identifies them or applies a diagnostic label. Intrinsic does not mean independent of context: learning, social coupling, and current conditions can constitute or change that organisation. A report or classification describes the structure under a measurement contract; it does not confer the structure.

Practical recovery is an inverse problem. Similar observed choices can arise from different apertures, acceptable regions, weights, beliefs, or constraints, and a self-description need not identify the operative profile. Matched comparisons may narrow the compatible family without recovering one unique shape, as in the conditional reconstruction of Eq. (18). The distinction between a defined structure and uncertain access to it is therefore retained: uncertainty about the profile belongs in the measurement, without making the defining properties dependent on an observer's judgement.

A general account of psychological differences would be a separate programme. Many clinical categories are operationalised through symptoms, distress, or impairment; NIMH's RDoC framework instead investigates underlying functional dimensions and mechanisms that can account for symptoms (National Institute of Mental Health, n.d.). This is close prior work for the proposed direction. In the present proposal, felt difficulty is a modelled assessment of an operative state or transition. The intrinsic definitional object is the operative organisation, independently of whether its model assesses that organisation as difficult or problematic. Such assessments can provide evidence and causally change subsequent conduct; they need not identify the underlying profile. The relevant distinction is between diagnostic recognition criteria and a defining account of operative organisation. The present paper uses ethical-aperture differences to illustrate consequence accounting. It does not develop a general psychological classification or establish that its structural profile replaces a clinical diagnosis.

The actual consequence account remains distinct from this aperture. A victim's lost continuation does not become smaller because an actor represents it poorly or gives it little weight. Conversely, reduced self-censure, aversion, or identity conflict can be a real difference in the actor's own consequences, where grounded in their organisation. This is a difference in attributed cost, in addition to any difference in its evaluation. It connects directly to the state-dependent operator account in Eq. (16).

For a conditional representation, let o_j be the information available to actor j and Q_j its processing into admitted features. The represented features are $\widehat{z}_j = Q_j(o_j)$, and a stated linear decision model applies $w_j \cdot \widehat{z}_j$. Different input apertures or Q_j can produce different admitted accounts from the same available information. Different ethical-region predicates can admit different continuations within a common represented manifold; different w_j can produce different choices within the same admitted region. Different actual actor states can also produce different internal burdens. These are separately investigable routes. A difference in behaviour alone does not identify which route changed, nor demonstrate that every difference is a fixed weighting. Attention, learning, impulsivity, and policy stability can require additional state or decision terms.

Relevant research distinguishes abstract judgement from hypothetical choice. In a study of 102 students, Tassy and colleagues found that elevated psychopathic traits predicted more sacrificial, outcome-favouring answers to hypothetical choice questions, without the corresponding association for acceptability judgements (Tassy et al., 2013). The result concerns self-report traits and imagined choices, not enacted harm. Hauser and colleagues studied 60 male offenders and found distinct associations by psychopathy facet and emotion-processing condition (Hauser et al., 2021). Conditions also used different dilemmas and were not randomised, limiting causal attribution to emotion processing. These findings support separating explicit judgement, processing, and choice, while qualifying a uniform profile. They do not recover the framework's aperture geometry, acceptability predicates, or weights, and do not establish absent ethics, zero

empathy, or zero internal cost for every person with elevated traits.

Under this interpretation, differences in ethical application are differences within the informational and decision machinery already specified. The account can register the actor's smaller or different internal burden alongside unchanged external loss. It does not require an extra category of existence or an eligibility exception for those affected. This is an explanatory application of the framework, with its empirical mechanisms still to be identified under matched conditions.

8.5 Recognition can cause further cost without creating the original cost

Observer independence does not require every consequence to precede awareness. An external recipient's lost future can exist before anyone knows about it. Later learning about the event can transform an actor or witness, creating additional costs in that system. The information received is then part of the causal transition, rather than a passive observer conferring ethical value. Proposition 5.1 holds for the fixed causal case; changing what information reaches a system changes that case.

A false belief can therefore have real consequences for its bearer even when the supposed external event did not occur. Record the represented event's causal effects in the bearer and distinguish them from the uninstantiated external injury. Conversely, an external loss is not removed because no actor feels guilt. Guilt is not an accounting instrument that certifies or nullifies all other costs.

Repair has the same separation. Restoring an external recipient's capability does not automatically restore the actor's self-relations, a witness's trust, or institutional coordination. Each requires its own grounded restoration condition and feasible repair paths. Nor does later repair cancel the intervening path account. Whole-system accounting makes these different bills visible in one typed family while keeping their incidence and units distinct.

9 Operative valuation and a self-model that rejects it

Consider a self whose decisions already implement a numerical valuation or its functional equivalent, while its self-model rejects the explicit account of that valuation as uncomfortable. It can hold that a human life cannot have a value assigned while its operative choices still make trade-offs involving lives. Writing an actual number exposes the valuation to the self-model. The distinction is between what the self carries and what its model of itself admits.

Paper 01b supplies this distinction: the self-model is a proper, incomplete part of the self, and its realisation account distinguishes changed modelling from newly occurring facts (Section 6, Remark 6.3). An operative valuation need not be stored as a literal numeral. It can be realised through relative weightings, thresholds, or an action ordering. The present hypothesis concerns a valuation already operative in those relations. Explicit acknowledgement need not create it.

The closest prior work includes taboo trade-offs, sacred values, and motivated reasoning. Fiske and Tetlock examine why some explicit exchanges are offensive in themselves (Fiske and Tetlock, 1997). Tetlock and colleagues studied responses to monetising protected values (Tetlock et al., 2000), and later work discusses shadow prices under scarcity (Tetlock et al., 2017). Kunda's account describes motivation shaping access to and evaluation of beliefs (Kunda, 1990). These supply relevant mechanisms and observations. The specific explanation proposed here is that discomfort can protect a self-model from admitting an already operative valuation. Discomfort alone is not proof that this explanation applies in a particular case.

9.1 Models of an agent have no automatic accuracy hierarchy

Can another person know an agent better than the agent knows themselves? For specified aspects of the agent, the answer can be yes. Paper 01b's self-model account distinguishes access mode from accuracy: the self-model is a proper, incomplete part of the self, while an external model

has its own access, resolution, and modelling investment (Proposition 6.1 and its subsequent discussion). Neither position alone establishes greater accuracy.

Impaired self-awareness supplies a concrete case. Cocchini and colleagues documented unawareness of motor impairment after left-hemisphere brain damage, using neurological assessment, self-assessment, and caregiver ratings (Cocchini et al., 2009). Observed impairment can conflict with the person's account of their own ability. The study also illustrates that the assessment method affects what can be recovered: language-dependent interviews and a less language-dependent instrument identified different levels of unawareness. This concerns particular impairments and assessments, without making disability itself a general loss of self-knowledge.

Self-modelling and modelling other agents are closely related capabilities. Both construct selective representations and can use shared skills in distinction, inference, prediction, and calibration. Their embedding differs: a self-model is an operative part of the organisation it models, so revising it can change its own target's subsequent dispositions and continuations. An external model has its own access and affects the target through whatever coupling is instantiated. Skill in one can support skill in the other, while their different access and embedding prevent accuracy in one from guaranteeing accuracy in the other.

Domain expertise provides a further illustration. A master may recognise a junior's likely move, mistake, or response from a familiar configuration before the junior can predict or articulate it. The master's learned structure can resolve domain distinctions and continuation patterns that the junior's self-model does not yet hold. The comparison concerns predictive accuracy within that domain under matched information and conditions, assessed against the junior's subsequent actions. It need not imply broader access to the junior's experience or greater accuracy about the whole person. This is an illustration of how modelling skill can affect comparative accuracy, rather than a new empirical result of this paper.

Every model of an agent is therefore examined through its own aperture and calibration, including the agent's self-model, another person's model, and an institution's model. Aperture specifies represented dimensions, resolution, and horizon; accuracy concerns agreement with grounded observations or predictions under declared comparison conditions. A model with broader coverage can be poorly calibrated, while a narrower one can be accurate within its scope. Models can also be accurate on different aspects without one being uniformly better. Accuracy comparisons require aligned target properties, conditions, and a stated criterion.

For ethical inquiry, self-report is evidence about a model of the operative self, rather than a definition of the self's actual capabilities or dispositions. External models can improve that account where their comparative accuracy is grounded. They can also omit or misrepresent it. This preserves the intrinsic definitional object while allowing variable access to it, and explains why recovering an operative ethical profile can require comparisons among several models instead of accepting one vantage as definitive.

9.2 Recovering the operative self from choices

A fundamental epistemic problem of ethical inquiry is the relation between the self and its self-model. The working hypothesis is that much ethical inquiry attempts to reconstruct the self's operative dispositions from revealed choices, and to calibrate its self-description against that evidence. Here "recovering the self" means recovering aspects of its consequential organisation; finite observations need not identify the entire self or a unique internal mechanism. The self-model is itself active within that organisation: its commitments can change later choices, rather than merely report an otherwise fixed actor.

Revealed-preference theory supplies an established method for inferring comparisons from choices. Samuelson's original construction compares a selected bundle with an alternative that was affordable under the same conditions (Samuelson, 1938). The present application compares inferred operative dispositions with the actor's admitted self-description. It does not require a literal numerical representation inside the actor.

For a conditional linear example, let observation k record the selected action a_k, the alternatives A_k actually available and known to the actor, and aligned feature vectors $z_k(a)$ under the actor's outcome beliefs. Let W_0 be a declared family of candidate weights, with units and any normalisation specified. For informative weight reconstruction, exclude the zero vector, which otherwise makes every action tie, and declare a scale convention if a unique numerical magnitude is sought: positive rescaling preserves these choice inequalities. Assume that one stable weight vector minimises the score in all recorded comparisons, allowing ties. The compatible operative models are

$$W_{\mathrm{op}} = \{w \in W_0 : \ w \cdot z_k(a_k) \leq w \cdot z_k(a) \quad \text{for every } k \text{ and } a \in A_k\}. \tag{18}$$

On a finite candidate family this is an implementable filter using `evaluate`. Equation (25) below is its one-conversion case. Additional comparisons can shrink the compatible family while the model and conditions remain fixed. An empty family rejects the stated candidate family or choice assumptions; several surviving models leave the reconstruction underdetermined. A stochastic or changing policy requires a correspondingly different inference model.

Feasibility, knowledge, coercion, outcome beliefs, and context matter to this inference. An unavailable alternative supplies no rejected-choice evidence. Features inferred from an actor's beliefs need not equal the measured actual consequences: both belong in the account with their distinction retained. Revealed preferences help reconstruct how the actor responds; the affected systems' losses do not depend on that reconstruction. Nor does discovering an operative disposition establish that it ought to be retained. Inquiry can calibrate the self-model, revise the operative disposition, or change the conditions of action. These are distinct interventions with calculable effects.

9.3 Energy-state management and deformation of decision fields

Within the proposed ontology, the foregoing processes concern states and transitions in an n-dimensional relational manifold. Relevant dimensions can include resources, enabling relations, affective burdens, self-model commitments, social sanctions, and the continuations available to affected frames. Paper 02 supplies the operational manifold, local transition costs, and deformability of the fields over it (Definitions 6.1, 12.1, and 12.6). The self and its model are part of the same evolving organisation, so changing the model is itself a transition in that manifold. It can also deform the decision field governing subsequent transitions.

For a conditional scalar decision model, write

$$\Phi_M(a \mid x) = w_M \cdot \widehat{z}_M(a \mid x), \tag{19}$$

where x is the current joint state, $\widehat{z}_M$ records the action features admitted by the model, and w_M supplies declared conversions into the score's unit. Under a stated minimising policy, reducing an action's score relative to its alternatives can change which transition is selected. A change in represented loss, self-censure, anticipated social sanctions, or expected benefit can all change this ordering. Energy and resource quantities retain their grounded units; a single scalar decision field exists only after its conversions are specified. Capacity loss remains a separately typed consequence even where energy is spent to produce or prevent it.

This supplies a common geometry for apparently different ethical moves. Admitting an uncomfortable valuation can require a costly self-model transition. Keeping the valuation unadmitted can preserve an existing organisation while constraining later modelling. A forced comparison can make that organisation incompatible with the requested commitment. Reclassifying an affected group can lower the represented cost of harming its members. Each is a possible way of changing states or transition prices in the manifold. The claim that organisms implement these moves through compressed regulation is the paper's explanatory hypothesis, rather than a recovered neural energy model.

The two continuations "act under the current model" and "change the model, then act" must therefore both be accounted for. The second includes the price of changing the field: propaganda, social coordination, and internal revision need not be free. Once a category rule is operative, it may apply across many individuals without a separate confrontation with each person's relations. Whether that makes the complete course cheaper for a particular actor requires the declared account. Making a subsequent action cheap in a decision field does not imply that its affected frames lose fewer futures.

9.4 Play as an elicitation of otherwise unspoken comparisons

Games such as "fuck, marry, kill" and "would you rather" provide familiar illustrations of deliberately eliciting comparisons. The first assigns three people to sharply different imagined relations; the second requests a choice between specified alternatives. The first compares assignments of relations, which need not give one total ranking of the people across those relations. A participant can discover an ordering they had never made explicit, or encounter a comparison that their self-description resists acknowledging. The proposed mechanism is a forced local collapse of self-model ambiguity: answering commits the participant to a comparison that could previously remain open. In this series' terms, collapse resolves a particular distinction under a constraint; it need not destroy the self or settle every aspect of its model. The game makes continued noncommitment incompatible with supplying the requested answer.

Let D_M be the descriptions the self-model leaves open before the question, and let $D(q)$ be those compatible with an answer q under its literal reading. A self-model that admits the answer as evidence restricts its open descriptions to $D_M \cap D(q)$. A proper nonempty subset records a local resolution of ambiguity. An empty intersection exposes a conflict requiring revision of the prior description or of the admitted meaning of the answer. The forced answer creates a commitment to confront; psychological admission of that commitment remains a further event. A participant can refuse, joke, or reject the answer's application to themselves. Those responses locate where the attempted collapse was resisted, rather than proving that the implied self-description was admitted.

An ordering is already a functional equivalent of valuation for those alternatives, although it need not identify a cardinal number or a stable ranking in other contexts. The mechanism hypothesis is that playful forced choice can make an operative comparison explicit and pressure the self-model to resolve its account of that comparison.

These are hypothetical elicited preferences. An answer is an actual choice of what to say in the game, with possible social and self-modelling consequences; it is not an observed performance of the imagined actions. Humour, audience, provocation, refusal, and the imposed options can shape the answer. The examples therefore illustrate a method of confronting the self-model with comparisons, without independently validating the proposed psychological mechanism. A study would distinguish answers, admission as self-description, and consequential choices under matched conditions. The game itself can also change the actor, so later answers need not simply uncover a valuation unchanged by elicitation.

9.5 Refusing a possible action versus judging it unlikely

Statements of surprise in crime programmes provide another illustration: "I cannot imagine someone like that doing this." The proposed interpretation is that some such responses express refusal to admit a reachable action into a model of oneself or another person, rather than a prior estimate that the action had low probability. The observed statement alone does not identify that mechanism. Surprise can also reflect a reasonable low-probability estimate, lack of relevant knowledge, or the social expression of condemnation.

The formal distinction concerns the model's represented possibilities. Let $\widehat{R}_M$ be the action continuations admitted by a self-model or a model of another actor, and let R_θ be the

independently grounded reachable continuations under the declared conditions. A particular harmful continuation $h \in R_\theta \setminus \widehat{R}_M$ is a model omission. If $h \in \widehat{R}_M$ and the model assigns it a small probability, it is instead a represented but unlikely event. If no probability has been assigned, omission is not a measured probability of zero. The difference is between which alternatives the model admits and how it distributes expectation among admitted alternatives. Establishing that an omission is motivated requires evidence beyond establishing the omission itself.

A model of "people like us" can support a person's own self-description. Evidence that an apparently ordinary person performed a harmful action can therefore require revision both of the model of that person and of the observer's account of their own possible conduct. The protective-self-model hypothesis developed here predicts that some resistance occurs at this admission step; compression alone does not require resistance to updating. Recognition makes a previously excluded continuation available to modelling; it need not make that continuation more reachable in the original event. Recognition can nevertheless change the observer's later actions and relationships, whose consequences belong in the whole-system account.

Bandura's account of moral disengagement is relevant prior work: it describes self-regulation and mechanisms for preserving a moral self-description while performing harmful actions, including justification, displacement of responsibility, and minimisation of harm (Bandura, 2002). This supports examining interacting personal and social conditions rather than assuming a moral identity fixes conduct across circumstances. It does not establish that every individual can perform every crime, or diagnose why a particular viewer expresses surprise. The claim needed here is narrower: self-description alone cannot exclude a continuation whose feasibility has been independently grounded. Reachability is conditional on actual capability, access, resources, and circumstances. Possibility also does not establish a high probability, willingness, or a uniform propensity across people.

9.6 Dehumanisation and exclusion from the represented account

Two changes of the decision field should be distinguished. An actor can retain an entity's represented losses and regard violence against it as deserved or justified, giving the action sufficient favourable significance in the evaluation. Alternatively, the actor can lower the represented standing of the entity so that harming it contributes little loss to the decision field. The proposed interpretation of the genocidal reclassification move concerns this second route: reducing the represented cost of harming members of a category, without requiring a separate judgement that each deserves punishment. The routes can coexist in the same actor or institution.

An actor who regards killing a person as costly can reclassify a group as non-persons, pests, or threats, and thereby remove or attenuate the losses admitted by their own decision model. Group classification can apply that exclusion across many individuals without confronting each person's continuing relations. The proposed mechanism concerns a reduction in represented loss and in restraints on action; it does not reduce the affected person's informational loss merely by changing a label.

For an illustrative binary exclusion rule, let $e_{i,M} \in \{0, 1\}$ specify whether an actor's model admits an affected frame's loss entry. For a fixed finite grounded loss L_i, its represented entry would be

$$\widehat{L}_{i,M} = e_{i,M} L_i. \tag{20}$$

Changing $e_{i,M}$ from one to zero erases that entry from the actor's represented account while leaving L_i fixed. This describes a gate in the actor's model; it is not an eligibility rule of the present framework. Partial attenuation, changed weights, and reinterpreted outcome beliefs provide other ways of changing an operative evaluation. The person whose future is destroyed

remains an affected frame in the whole-system account regardless of the perpetrator's category assignment.

Kelman's account of sanctioned massacres is particularly close to this interpretation: it examines authorisation, routinisation, and dehumanisation as processes weakening restraints on violence against groups that have not themselves engaged in hostile actions (Kelman, 1973). Landry and colleagues' analysis of Nazi propaganda found decreasing attribution of mental experience to Jews before the Holocaust, consistent with disengagement of moral concern (Landry et al., 2022). The latter measures propaganda language, not the distribution of individual perpetrators' decision mechanisms; its observational findings and missing archival data do not establish causal sufficiency or a majority prevalence. The predominance of the proposed route across perpetrators remains an empirical question.

Bandura identifies dehumanisation and victim blaming among mechanisms for selective moral disengagement (Bandura, 2002). The proposed informational reading is that reclassification can permit a moral self-model to remain compatible with harmful action by excluding consequences or changing their interpretation. Dehumanisation is not a necessary explanation for every atrocity. Rai and colleagues distinguish instrumental violence from violence perceived by its perpetrators as morally justified; their experiments found increased willingness for instrumental harm under dehumanising descriptions, without the same increase for morally motivated harm (Rai et al., 2017). These studies concern attitudes and vignette responses, not a complete causal demonstration of genocide. A perpetrator can recognise a victim's humanity and still construe harming that victim as deserved. Both exclusion and justification can be analysed as changes to the model or its evaluation, with their actual consequences retained.

The reclassification process also has consequences of its own. It can reduce real self-censure or social sanctions for perpetrators, while propaganda, isolation, institutional exclusion, and organised violence change victims' and witnesses' actual enabling relations and futures. These effects must be counted where grounded. The framework therefore permits an actor's burden to fall while another frame's loss rises. A saving in one attributed dimension does not erase a loss in another. Relabelling with victims' capabilities and continuations held fixed does not change their grounded loss. Effects of that relabelling on actors, institutions, or other affected frames can still change the whole-system account, even when the victims' measured loss remains fixed.

9.7 Research support for context-sensitive ethical application

The proposed mechanism has relevant experimental and historical support. These findings support context-sensitive choices and changes in admitted consequences; their interpretation through the present account remains a hypothesis. They do not independently establish a population distribution of intrinsic moral character or identify the brain's implementation.

Dana and colleagues experimentally reduced transparency between choices and recipients' outcomes in a binary dictator game. Generosity fell when actors could leave consequences uncertain, even though a fair outcome remained available (Dana et al., 2007). This is relevant to preserving a favourable self-description while relaxing practical constraints on selfish action.

Vu and colleagues' meta-analysis covered 6,531 participants and 56 treatment effects. It reported approximately 40% avoiding easily obtainable information about consequences for others, and a 15.6-percentage-point reduction in altruistic behaviour when the opportunity for ignorance was available (Vu et al., 2023). The result supports information avoidance as a mechanism for changing conduct. Avoiding new outcome information does not, by itself, demonstrate refusal to acknowledge an already operative valuation; those are distinct evidential claims.

In genocide research, Luft's analysis of interviews with 31 Hutu documents behavioural crossing between killing, desistance, and saving during the 1994 Rwandan genocide (Luft, 2015). This supports a dynamic account of conduct rather than fixed actor categories. The qualitative sample does not estimate a universal proportion of ambivalent people.

Luft's later analysis examines selected first-time killers whose initial cognitive and emotional difficulty diminished as killing became routine and victims became socially meaningless (Luft, 2023). The study triangulates retrospective perpetrator accounts with reports and survivor/rescuer testimony. It closely matches lowered represented relevance, while qualifying causal order: in the examined trajectories, dehumanisation followed participation and habituation. The change need not involve conscious reclassification or explicit self-talk. This supports possible feedback from action into subsequent decision fields, without proving that every perpetrator follows that trajectory.

The lack of explicit acknowledgement matters to the self-model hypothesis. Where an actor's admitted standards condemn a change in conduct, consciously admitting that change as their own disposition would require revising a favourable self-description: "I am becoming someone who does this." Implicit habituation can instead change operative responses and represented relevance while that description remains uncorrected. In the proposed account, this can avoid the additional burden of confronting a conflict between conduct and admitted identity. The actual self changes; its model need not admit the change as becoming a worse person. Bandura's account supplies closely related prior work on harmful conduct alongside preserved moral self-description (Bandura, 2002). Conscious reinterpretation can also preserve that description by presenting the conduct as justified. Thus consciousness alone does not entail self-condemnation. The specific prediction concerns acknowledgement of deterioration under the actor's own retained standards. Lack of explicit acknowledgement of that deterioration need not mean lack of awareness of the acts or their consequences. Whether avoiding this confrontation helps sustain a particular trajectory requires evidence beyond habituation alone.

Asch's conformity experiments supply adjacent evidence that social conditions change responses while leaving substantial individual resistance (Asch, 1956). Their task concerns perceptual judgements, not ethical choice. Asch also reports a non-normal distribution of observed error counts. Those counts neither establish nor refute a normal distribution of latent ethical margins; the quantities and tasks differ.

Together these findings give the framework empirically supported phenomena to explain: information-conditioned prosocial action, changing social relevance, behavioural switching, and pressure alongside resistance. They support susceptibility to changes in a decision landscape. They do not uniquely identify the self-model mechanism, demonstrate calibrated low margins for a majority, or validate the Gaussian assumption below. A direct assessment would measure choices and available alternatives under matched interventions, distinguish changes in information from changes in classification, and estimate the margin distribution instead of presupposing its shape. Habituation requires a changing-state model; the static common-shift calculation is one conditional slice of that wider process.

9.8 A population model of ambivalence and shifted thresholds

The proposed population hypothesis is that most people are ambivalent in their application of ethical constraints, rather than committed to destructive conduct. In the present geometry, a change in classification can therefore move ordinary actors across a decision threshold by lowering represented loss. This hypothesis concerns the population's response to the account. The derivation of occurring consequences does not depend on how the population is distributed.

For a calculable conditional model, fix one harmful action h and a comparison action b under stated conditions. For actor j, define the converted decision margin

$$m_j = \Phi_{M_j}(h \mid x_j) - \Phi_{M_j}(b \mid x_j). \tag{21}$$

Positive m_j favours the comparison action under the assumed minimising policy; negative m_j favours h. Use a common calibrated score unit and first let the margin distribution have any continuous cumulative distribution function F. A classification intervention reducing margins

by a common $\delta \geq 0$ gives $m'_j = m_j - \delta$ and hence

$$\Pr(m'_j < 0) = F(\delta). \tag{22}$$

The newly crossing fraction is $F(\delta) - F(0)$. Continuity removes tie mass; a distribution with atoms requires an explicit tie rule and the corresponding one-sided probabilities. The threshold-shift mechanism therefore does not require a normal distribution.

For the proposed normal case, assume $m_j \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$, and write F_{N} for the standard normal cumulative distribution function. This is a distribution of conditional decision margins, not a numerical measure of goodness or evil. Normality remains a modelling assumption rather than an empirical result established by the cited studies. The conditional proportion selecting h is

$$p_h(\delta) = \Pr(m_j < \delta) = F_{\mathrm{N}}\left(\frac{\delta - \mu}{\sigma}\right). \tag{23}$$

Thus $\mu > 0$ gives a baseline in which fewer than half select h, while a shift $\delta > \mu$ makes the same action favoured by more than half in the conditional model. The actors need not acquire a new commitment to cruelty: their represented comparison has changed. When $\mu > 0$, the admissible shift $\delta = \mu$ gives a predicted fraction of one half, and sensitivity to an additional common shift is greatest there. The calculation holds the available actions and other margin components fixed; coercion, heterogeneous shifts, changed opportunities, or social feedback require those features to be modelled separately. Only actors with $0 < m_j < \delta$ newly cross the threshold in this direction; their fraction is $p_h(\delta) - p_h(0)$. Actors already favouring h contribute to the resulting fraction but have not been recruited by this shift.

To make "ambivalent majority" inspectable, declare a margin width $\varepsilon > 0$ and use $|m_j| \leq \varepsilon$ as a proxy for choices readily changed by a shift of comparable size. Its population fraction is

$$p_{\mathrm{near}} = F_{\mathrm{N}}\left(\frac{\varepsilon - \mu}{\sigma}\right) - F_{\mathrm{N}}\left(\frac{-\varepsilon - \mu}{\sigma}\right). \tag{24}$$

The proposed majority condition is $p_{\mathrm{near}} > 1/2$. It is a condition on μ, σ, and a substantively justified width, rather than a consequence of normality alone. A small net margin may also arise from strong opposing commitments; identifying psychological ambivalence requires further evidence. These quantities describe possible participation under matched conditions, without inferring universal criminal capability or intrinsic moral character. The reference function `normal_response` calculates the conditional fractions from declared parameters; it does not fit or validate them.

Granovetter's threshold models are direct prior art for aggregate changes arising from distributions of individual decision thresholds (Granovetter, 1978). His actors' thresholds depend on others' participation, whereas the calculation here holds the common shift exogenous. The informational contribution proposed here connects a classification-induced shift to the losses omitted from the actor's model, while retaining those losses in the whole-system account. A majority participating in the resulting action would not, by itself, establish either the assumed normal distribution or a majority that originally desired its consequences. In this mechanism, actors who actively deform the field can change the choices of a broad population with small margins, without each influenced actor acquiring an independent commitment to the destructive outcome.

9.9 Three different quantities

First, there is the consequence account of the proposed action: which continuing lives, relations, capabilities, and resources it changes. Second, there is the operative valuation: how the self actually weighs or orders losses and burdens. Its self-model can admit, omit, or reject a representation of that valuation. Third, there is the consequence account of requiring the

representation to become explicit. The life affected by the action, the number written by the actor, and the actor changed by writing it occupy different places in the account.

A number can express a grounded quantity, a decision conversion, or an estimate with uncertainty. The label "the value of a human life" does not specify which of these it represents. The present framework derives an affected person's consequence account under a contract; a monetary conversion of that account is an additional declared evaluation. This distinction gives the written number an inspectable meaning without making a universal scalar an untyped property of every life.

A self-model's declaration of pricelessness leaves several possibilities open: a genuine decision constraint, an unresolved conversion, a socially maintained commitment, or a compression used alongside finite practical choices. A declaration alone does not identify which one holds. Neither does every finite resource choice identify a stable hidden price: the inference requires a decision model.

9.10 Quantifying an implied conversion

For a simple conditional model, let an intervention avoid $d > 0$ units of a declared loss measure and require $c \geq 0$ additional resource units. Let its score difference from the baseline be $c - \lambda d$, where $\lambda \geq 0$ converts loss units into those resource units. With other effects held fixed, accepting the intervention implies $\lambda \geq c/d$ and rejecting it implies $\lambda \leq c/d$, allowing equality at a tie. Several choices therefore imply

$$\max_{\text{accepted}} c/d \ \leq \lambda \leq \ \min_{\text{rejected}} c/d. \tag{25}$$

An empty accepted set gives no positive lower bound; an empty rejected set gives no finite upper bound. Inconsistent bounds reject this fixed scalar model. The inference concerns the declared loss unit and these choices; it is not a recovered intrinsic price of a particular person.

For example, acceptance at three resource units per avoided loss unit and rejection at five give $3 \leq \lambda \leq 5$. Writing $\lambda = 4$ proposes a specific conversion compatible with those choices. It does not establish that four is the operative value; the choices identify an interval. A self-model can reject every finite member of the interval while the operative choices still delimit it. The writing does not alter the original recipient's already fixed consequences. It can alter the writer's organisation, their future choices, and the relations in which the written comparison is available. A policy constraint or nonlinear rule need not admit the scalar inference at all. The reference function `shadow_price_bounds` implements the finite interval calculation, including inconsistent inputs.

9.11 Quantifying what the self-model will not admit

Choose a declared finite grid H of candidate conversions. Let $\Lambda_{\text{op}} \subseteq H$ contain the values compatible with the operative choices under the stated decision model. Let $A_M \subseteq H$ contain the values the self-model explicitly admits as descriptions of that same operative valuation, under matched conditions. Asking whether a number describes the self's choices differs from asking whether the person endorses it as an ideal.

For nonempty Λ_{op}, define

$$E_M = \Lambda_{\text{op}} \setminus A_M, \tag{26}$$

$$g_M = \mathbf{1}\{\Lambda_{\text{op}} \cap A_M = \varnothing\}, \tag{27}$$

$$f_M = \frac{|E_M|}{|\Lambda_{\text{op}}|}. \tag{28}$$

The excluded set preserves which candidates are rejected. The indicator records complete incompatibility within the declared grid; the fraction is a uniform candidate-count summary

of rejection. It is not a measure of discomfort's intensity, nor a moral weighting of candidate values. Refining or resampling the grid changes this summary's contract.

For $H = \{3, 4, 5\}$ and operative bounds $3 \leq \lambda \leq 5$, rejection of all three as possible descriptions gives $E_M = H$, $g_M = 1$, and $f_M = 1$. The operation need not introduce new external facts. It forces the self-model to confront the valuation its own choices already constrain. If no candidate matches the choices, the grid or decision model has failed; the gap is undefined, not evidence that the self-model rejects a recovered valuation. If some candidates have not been probed, their admission status remains unknown. The reference function `valuation_gap` implements the finite comparison for fully classified candidates.

This is an operational quantification of the proposed distinction. It locates a mismatch between operative valuation and admitted self-description without identifying a unique brain variable. Explaining the mismatch specifically by discomfort requires independent evidence of aversion or avoidance, with audience, incentives, misunderstanding, and changed decision conditions distinguished. The cost of sustaining or resolving the mismatch is a further consequence account, rather than another name for the indicator.

9.12 Quantifying confrontation and acknowledgement

Let e_0 be the condition in which operative valuation remains unadmitted by the self-model and e_v the condition in which the actor confronts or writes an explicit candidate number v. Hold the compared external options, resource facts, and outcome beliefs fixed where the experiment permits. Apply the existing functions to

$$\mathcal{C}_\theta^{\text{number}}(v) = \mathcal{C}_\theta(e_v \mid e_0). \tag{29}$$

The condition e_v means confrontation or expression, not demonstrated admission. Someone can write a number while still rejecting it as a description of themselves. Acknowledgement requires a further observed change in A_M or other grounded evidence of changed self-modelling. For a verified admission condition e_{admit}, take the additional comparison $\mathcal{C}_\theta(e_{\text{admit}} \mid e_0)$; writing alone does not establish it.

This account records changes in the actor's and other affected frames' futures, access prices, expenditure, and course profiles. It can contain gained capability through clearer planning as well as lost trust or increased burden. No assumption of injury is built into the comparison.

Private reading, private writing, and publication are distinct conditions. Publication adds audiences and institutional consequences; requiring authorship can add responsibility or commitment. If the explicit condition also supplies new facts, changes incentives, or changes the audience, its account includes those effects. It cannot then be described as a pure effect of numerical format.

As an additional behavioural dimension, for aligned action distributions under matched circumstances x, define

$$B(v; x) = \tfrac{1}{2} \sum_u |\pi(u \mid x, e_v) - \pi(u \mid x, e_0)|. \tag{30}$$

This total-variation distance lies between zero and one and measures changed action probabilities. It is neither a moral weight nor a universal depth-of-knowledge score. Equal action distributions can coexist with changed self-relations, expenditure, or effects in another context. The function `policy_distance` computes a floating-point approximation with normalisation tolerance 10^{-12}; the exact bound applies to exactly normalised distributions. The existing functions compute Eq. (29) from the grounded branch models.

In this example, "knowing" versus "*knowing*" distinguishes carrying an operative valuation from admitting it into the self-model. The gap can already exist before a number is written. Acknowledgement changes the modelling; because the self-model is part of the self, that change

can also alter later action, self-relations, and expenditure. It need not change the historical consequences whose valuation was previously denied.

Moral injury is one possible consequence when confronting or revising this self-description damages enabling self-relations or trust. Acknowledgement can also improve calibration and make hidden costs accountable. The operative valuation, its rejection, and the cost of admitting or maintaining the rejection are separate objects within the whole-system account.

10 Compression and ethical behaviour

A calculable principle need not be explicitly calculated whenever a system acts. Organisms coordinate, avoid damage, and help others before acquiring a theory of ethics. The informational account proposes an underlying structure that can be compressed into affect, habits, expectations, and social practices.

Hypothesis 10.1 (Ethical behaviour as compression)**.** *A substantial part of human and animal ethical behaviour expresses compressed sensitivity to changes in affected capabilities, enabling relations, and burdens. The expression depends on available information, learned compression, and relative decision weightings. It need not represent a formal consequence account or reproduce the functions printed in this paper.*

Four objects must remain distinct: the actual consequence account $\mathcal{C}$, the information available to the actor, a compressed representation of that information, and the actor's decision weights. A poor report can hide a real cost. A coarse compression can merge different costs. Different weights can produce different responses to the same represented costs. These are different explanations and admit different corrections.

10.1 What compression can preserve

Let z be a finite consequence feature vector and let $Q(z)$ be a compressed record. For a fixed decision map d, exact preservation requires the decision to be recoverable from that record.

Proposition 10.2 (Exact decision preservation)**.** *There exists a map g with $d = g \circ Q$ if and only if d is constant on every fibre of Q: whenever $Q(z) = Q(z')$, then $d(z) = d(z')$.*

Proof. If $d = g \circ Q$, equal compressed records give equal decisions. Conversely, if d is constant on each fibre, define $g(q)$ as the common decision of the vectors mapping to q. The condition makes this definition well-defined on the image of Q. □

For a concrete finite example, suppose a system compares actions using Eq. (14) with weights $(1, 1, 2)$. Compression to the single feature $Q(z) = L_A + K_{B,r} - 2G_B$ preserves its score and all score-based choices, although it discards the separate effects. That compression does not generally preserve choices after the weights change. Preserving a decision is weaker than preserving the full cost account.

Approximate compression also has a precise boundary. Let $\widehat{z}(a)$ be a reconstructed feature vector and suppose $|w \cdot \widehat{z}(a) - w \cdot z(a)| \leq \epsilon$ for every feasible action. If $\widehat{a}$ minimises the reconstructed score and a^* the actual score, then

$$w \cdot z(\widehat{a}) - w \cdot z(a^*) \leq 2\epsilon. \tag{31}$$

Indeed, the actual score of $\widehat{a}$ is at most its reconstructed score plus ϵ; minimisation bounds that reconstructed score by the one at a^*; and reconstruction contributes at most a further ϵ. If component errors satisfy $|\widehat{z}_j - z_j| \leq e_j$, one admissible bound is $\epsilon = \sum_j |w_j| e_j$.

These results demonstrate how a principle can be expressed through compressed records, and what such expression loses. They do not establish that brains use this particular feature vector, compressor, or minimisation rule. The hypothesis concerns the informational principle expressed through behaviour, with neural implementation left open.

10.2 Evidence for ethical behaviour without explicit theory

Haidt's social intuitionist account places moral judgement partly before explicit reasoning (Haidt, 2001). Instrumental helping in young children and chimpanzees (Warneken and Tomasello, 2006), and rats avoiding an action that harms a neighbour (Hernandez-Lallement et al., 2020), show behaviours that do not require an articulated ethical theory. Their evidential role is to support the possibility of ethical action preceding ethical explanation.

The informational hypothesis adds a proposed organising principle for such behaviour. Helping can restore another's action possibilities; avoidance can prevent an imposed burden; reciprocity can preserve enabling relations. The studies do not directly measure the full continuation account. Their behaviour can also be explained through learning, affect, affiliation, and reward expectation. Distinguishing explanations requires predictions about changed actual futures, changed information, and changed weights.

Computational moral-choice research already fits relative sensitivities to outcomes. Studies of harms to oneself and others (Crockett et al., 2014) and of different moral strategies (van Baar et al., 2019) demonstrate implementable decision models. The contribution here is not the invention of weighted moral computation. It is a cost account derived before those weights, together with a compression interpretation of the resulting behaviour.

Animal paradigms also require care about which variable changes. Reward attainability can explain responses attributed to inequity in dogs (McGetrick et al., 2023). This is relevant to the present framework: an experiment that changes attainable outcomes must distinguish that change from a changed comparison weight. Behaviour alone does not identify which account or priority an animal uses.

10.3 Graded treatment of entities

Human moral consideration often varies with perceived organisation, agency, experience, persistent relationships, and similarity. Mind-perception research relates attribution of agency and experience to different aspects of moral judgement (Gray et al., 2007). Empathy and compassion across species also vary with evolutionary divergence (Miralles et al., 2019). These findings are compatible with coarse proxies for informational organisation, but they do not directly measure information-defined cost.

The hypothesis explains such proxies as compression: visible similarity, familiarity, responsiveness, and continuity can make another's future structure easier to represent. The proxy can be informative, biased, or misleading. A familiar agent with few affected continuations and an unfamiliar agent with many can reverse the ranking suggested by appearance. The account remains tied to the actual relations while the actor's representation can fail to track them.

Food choices illustrate the distinction. People can give different priorities to ending an animal's continuing organisation, producing milk or eggs, harming plants, and preserving their own nutrition or cultural practices. Vegetarianism and veganism therefore need not express identical weights or identical causal accounts. A YouGov survey reported more self-identified vegetarians than vegans in Great Britain and the United States, with different margins (Nguyen, 2024). This is contextual evidence, not a universal ordering of diets or a measurement of the animals' informational cost.

The proposed trend is greater consideration for entities represented as bearing richer consequential organisation and relationships. It is not a rule that humans consistently maximise that quantity, a claim that food practices are uniform, or a hierarchy of reality. Its empirical test must separate perceived structure, actual affected futures, and competing weights. The game case makes that separation unusually inspectable because the underlying continuation model can be available.

11 Relation to earlier work

The ingredients have substantial predecessors. The comparison concerns what each theory takes as given and what it calculates, rather than a claim that information has only now entered ethics.

11.1 Information ethics and cybernetics

Floridi's information ethics extends ethical consideration beyond a human-centred population (Floridi, 1999). Floridi and Sanders connect evil to impoverishment of informational structure and state duties concerning entropy (Floridi and Sanders, 1999). Their entropy is not simply raw Shannon uncertainty. The close overlap is an informational, broadly ontological basis for ethical consideration. The present proposal develops attributed continuation and expenditure functions while placing priorities in a subsequent decision rule; it does not import a duty to maximise every entity's informational state.

Von Foerster connects ethics with increasing available choices (von Foerster, 2003). Ashby's variety supplies an established relation between available distinctions and regulation (Ashby, 1956). The present derivation calculates which options and enabling conditions change. A choice-increasing imperative would be a further evaluation rule over that account. Causal entropic forces likewise demonstrate dynamics organised around future path entropy (Wissner-Gross and Freer, 2013); the adopted maximisation principle is not by itself a cross-agent cost allocation.

11.2 Empowerment and reachability-based impact

Empowerment gives an operational information-theoretic measure of control. Salge and Polani explicitly use human, robot and transfer empowerment in a proposal for robotic ethics, including weights and human priority (Salge and Polani, 2017). This is a close and implementable predecessor. The present account retains attributed losses, gains, expenditures and courses without imposing a human-serving mission as a condition of its calculation.

Relative reachability compares side effects against a baseline (Krakovna et al., 2018). Attainable utility preservation measures changes in capacities to optimise auxiliary rewards (Turner et al., 2020). Both precede this paper's use of future capability as an impact measure. Their baseline and proxy concerns apply here: a convenient baseline can conceal a burden, and a proxy can conceal substitution. Retaining outcome identity, recipient incidence, and path effects exposes those omissions; it does not make every finite model complete.

Kolchinsky and Wolpert derive semantic information through correlations causally necessary for maintaining a system's existence (Kolchinsky and Wolpert, 2018). Their viability criterion connects informational significance to dynamics across living and nonliving systems. This paper instead compares affected continuations and enabling expenditures under declared contracts. Sen's capability approach also centres what an agent can actually do (Sen, 1980); its evaluative account supplies another important point of comparison without being the source of the finite set derivation.

11.3 Relation to the ethics already present in this series

The earlier papers already contain parts of the account. Paper 01 locates expenditure and attributed harm in affected frames. Paper 01b's communication analysis distinguishes structural influence from goal-relative beneficiary comparison. Its requirement for represented preferences concerns the latter comparison; it is not an eligibility condition for the capability losses here. Paper 02's reversibility preference (Proposition 16.2) is conditional on valuing the retained optionality. The current paper preserves that condition as a decision claim.

Paper 02b makes restoration without cancellation of an intervening course explicit. Paper 04 supplies consequential measure and prohibits an ungrounded sum across frames. Paper 09 limits

an institutional application rather than claiming a complete ethics; Papers 10 and 10a apply named objectives and priorities to particular futures. Those priorities can specialise the present decision layer. They are not thereby derived as universal weights.

Paper 11 assembles these distinctions into a consequence account with explicit functions, proves its conditional invariances, and adds the biological compression hypothesis. Its novelty claim rests on this arrangement and the work it does. Neither information ethics nor implementable moral models begin with this paper.

12 An application to AI alignment

Alignment is an adjacent application. Instruction-following systems trained with human feedback use judgements to shape a reward or training objective (Ouyang et al., 2022). Constitutional AI uses stated principles to guide feedback and revisions (Bai et al., 2022). In these representative approaches, feedback or principles enter the training criterion. They do not by themselves supply the independently attributed cost construction developed here.

A system could instead maintain a consequence account across candidate actions, estimate uncertainty, and expose the effects before its deployment weights select a policy. Human priorities, legal constraints, or institutional rules could then be explicit decision conditions. Changing those conditions would not rewrite the measured closure of another frame's options.

This application does not characterise all alignment research as preference fitting: empowerment, relative reachability and attainable utility preservation already overlap with the present machinery. Nor does a better account solve causal modelling, boundary choice, deceptive reporting, or conflicting goals. It offers a basis on which those conflicts and omissions can be made inspectable. An agent instructed to minimise a coarse reported cost can still exploit what the report leaves out.

13 Scope, challenges, and conclusion

The paper's central result is an implementable whole-system account of informational consequences whose occurrence does not require evaluative recognition. Its strongest claim is that a bearer need not first be approved as morally eligible: where the relevant consequence is instantiated, it is available to the accounting procedure. Its strongest boundary is that measurement still requires a grounding. The account can be wrong without the consequence waiting for a correct observer.

Four challenges specify where the theory can fail or require revision.

1. **Grounding and completeness.** A validated consequence that the proposed continuation, resource, and path construction cannot represent would expose a missing dimension. An outcome quotient that merges it would expose an inadequate resolution, rather than establish zero cost.

2. **Attribution and continuity.** A population or resource partition that loses an affected recipient, falsely identifies a replacement, or charges a shared expense twice invalidates that account. Nested collective accounts require explicit overlap treatment.

3. **Mathematical invariance.** Failure of the stated invariances under their actual hypotheses would invalidate a structural result. Altering the causal relations or measurement contract falls outside those hypotheses and must be reported as a changed comparison.

4. **Compression adequacy.** The biological hypothesis would lose explanatory value if changes in represented capabilities, enabling relations and burdens systematically failed to predict behaviour after competing information and priorities were controlled. Compression and weight

specifications must be fixed across the cases used to test a prediction. Arbitrarily refitting them for every observation is not an informative confirmation.

The proposed account of the ethical frame and the biological interpretation remain open to challenge. A rival explanation can dispute how closely moral dispositions track these consequences or identify additional causal organisation the account omits. Those are claims about coverage and explanation, rather than requirements that an observer confer value before a restriction or expenditure can occur. Moral injury sharpens this boundary: external benefit and internal damage can coexist, and a favourable decision score cannot certify the absence of either.

The derivation supplies separate functions for lost and gained futures, incremental burdens, loss and gain time profiles, and repair. A deterministic specialisation connects available distinctions to information capacity. Decisions then apply relative weights. Compression can preserve or approximate those decisions under stated conditions, providing a viable bridge to ethical behaviour before explicit ethical reasoning. Recovering the precise operations of a brain is not required for that bridge or for the account's use.

A future removed from an affected relation is removed whether it is seen, reported, or valued. That is the proposed lens shift. Whole-system accounting begins from what changed and who bears it, including the actor and the relations that sustain its continuations. Ethics becomes an accounting frame for consequences that narrower accounts exclude. The ontology supplies their place in the system; studying ethics exposes omissions and the compressed responses to them. The value placed on responding is a further operation.

A Finite reference functions

The following standard-library Python functions implement the finite specialisation. Graph edges encode admissibility; outcomes encode consequential identity. A budget-limited graph must carry its remaining budget in states. The reference functions do not discover the causal model, infer a moral population, or replace unknown inputs with zero.

For the protection case in Section 7.1, the calls are:

```
b = {"A": {1, 2, 3, 4, 5, 6}, "B": {"x"}}
a = {"A": {1, 2, 3}, "B": {"x", "y", "z", "u"}}
record = consequence_account(b, a, {}, {("B", "resource"): 1})
z = {"A_loss": record["frames"]["A"]["loss"],
     "B_burden": record["resources"][("B", "resource")],
     "B_gain": -record["frames"]["B"]["gain"]}
w = {"A_loss": 1, "B_burden": 1, "B_gain": 2}
# record: A loss 3; B gain 3; B resource difference 1.
# evaluate(z, w) = -2, as obtained in the text.
```

For the operative-valuation example of Section 9, the finite calls are:

```
bounds = shadow_price_bounds([(1, 3, True), (1, 5, False)])
# Bounds: lower 3, upper 5, consistent True.
candidates = {3, 4, 5}  # Declared finite grid inside those bounds.
gap = valuation_gap(candidates, set())
# Known rejection of every candidate: incompatible True,
# rejection_fraction 1.0, excluded {3, 4, 5}.
# These numbers measure model compatibility, not discomfort intensity.
```

These displayed results follow by direct substitution in the defined functions. The appendix supplies a reference implementation, not a reported software benchmark or a biological implementation.

```
"""Finite reference functions for Relational Information Accounting, Paper 11.

States and outcomes are hashable. Edges encode admissible transitions.
```

```
Encode clocks, remaining budgets, and identity in states when they matter.
A missing outcome (None) contributes no continuing outcome for that frame.
No evaluator weight appears in reachability or the consequence account.
Arithmetic overflow raises an error rather than reporting a measured infinity.
"""
from heapq import heappop, heappush
from itertools import count
from math import isclose, isfinite, log2
from statistics import NormalDist


def _finite_result(value, quantity):
    """Reject arithmetic overflow without confusing it with a model result."""
    if not isfinite(value):
        raise ValueError(f"{quantity} exceeds finite arithmetic range")
    return value


def reachable(edges, starts, horizon, outcome, *, exact=True):
    """Distinct outcomes at exactly horizon steps, or up to that horizon.

    Terminal persistence requires explicit self-loops in edges. Transition
    nondeterminism represents possible reachability, not assured control.
    Finite adjacency iterables are materialised for repeated visits.
    """
    if not isinstance(horizon, int) or horizon < 0:
        raise ValueError("horizon must be a nonnegative integer")
    edges = {state: tuple(steps) for state, steps in edges.items()}
    frontier = set(starts)
    visited = set(frontier)
    for _ in range(horizon):
        frontier = {v for u in frontier for v in edges.get(u, ())}
        visited.update(frontier)
    states = frontier if exact else visited
    return frozenset(y for s in states if (y := outcome(s)) is not None)


def compare_futures(baseline, action, masses=None):
    """Aligned outcome identities; optional nonnegative grounded masses."""
    baseline, action = frozenset(baseline), frozenset(action)
    lost, gained = baseline - action, action - baseline
    def measure(items):
        if masses is None:
            return len(items)
        values = [masses[x] for x in items]
        if any(not isfinite(v) or v < 0 for v in values):
            raise ValueError("outcome masses must be finite and nonnegative")
        return _finite_result(sum(values), "outcome mass")
    return {"lost": lost, "gained": gained,
            "loss": measure(lost), "gain": measure(gained)}


def resource_difference(baseline, action):
    """Keys are (payer, resource unit); absent keys mean measured zero."""
    keys = baseline.keys() | action.keys()
    values = list(baseline.values()) + list(action.values())
    if any(not isfinite(v) or v < 0 for v in values):
        raise ValueError("expenditures must be finite and nonnegative")
    return {k: action.get(k, 0) - baseline.get(k, 0) for k in keys}


def consequence_account(baseline, action, resources_b, resources_a,
                        masses=None):
```

```
    """Frame IDs map to measured reachable sets, including extinct IDs.

    An absent ID means a known empty set, never an unmeasured frame.
    Use a common grounding for outcome masses within each frame.
    """
    ids = baseline.keys() | action.keys()
    accounts = {i: compare_futures(baseline.get(i, ()),
                                   action.get(i, ()),
                                   None if masses is None else masses[i])
                for i in ids}
    return {"frames": accounts,
            "resources": resource_difference(resources_b, resources_a)}


def _change_profile(baseline_series, action_series, durations, masses, kind):
    if not (len(baseline_series) == len(action_series) == len(durations)):
        raise ValueError("series must have aligned intervals")
    if any(not isfinite(dt) or dt < 0 for dt in durations):
        raise ValueError("durations must be finite and nonnegative")
    profile = [compare_futures(b, a, masses)[kind]
               for b, a in zip(baseline_series, action_series)]
    exposure = sum(dt * value for dt, value in zip(durations, profile))
    return profile, _finite_result(exposure, f"{kind} exposure")


def loss_profile(baseline_series, action_series, durations, masses=None):
    """Piecewise constant loss, with a shared clock and interval lengths."""
    return _change_profile(baseline_series, action_series, durations,
                           masses, "loss")


def gain_profile(baseline_series, action_series, durations, masses=None):
    """Piecewise constant gain; keep separate from loss and its exposure."""
    return _change_profile(baseline_series, action_series, durations,
                           masses, "gain")


def deterministic_capacity(outcomes):
    """Capacity for distinct attainable outputs of a deterministic channel.

    The caller establishes that each output is attainable by a legal input.
    Empty outputs describe no channel, rather than a numerical log(0).
    """
    size = len(set(outcomes))
    return None if size == 0 else log2(size)


def least_repair(edges, start, goals):
    """Minimum scalar repair expenditure, or infinity if unreachable.

    edges[state] contains (successor, cost). Costs must share one grounded
    unit and be nonnegative. The goal set specifies restoration conditions.
    An overflowing explored path raises an error; infinity means unreachable.
    """
    edges = {state: tuple(steps) for state, steps in edges.items()}
    for steps in edges.values():
        if any(not isfinite(c) or c < 0 for _, c in steps):
            raise ValueError("repair costs must be finite and nonnegative")
    goals = set(goals)
    serial = count()
    queue = [(0, next(serial), start)]
    best = {start: 0}
    while queue:
```

```
        cost, _, state = heappop(queue)
        if cost != best[state]:
            continue
        if state in goals:
            return cost
        for successor, spend in edges.get(state, ()):
            candidate = _finite_result(cost + spend, "repair path cost")
            if candidate < best.get(successor, float("inf")):
                best[successor] = candidate
                heappush(queue, (candidate, next(serial), successor))
    return float("inf")


def evaluate(features, weights):
    """Explicit weighted decision score, following consequence calculation.

    Keys agree exactly; losses and resource burdens have positive feature
    signs, gains negative signs. Weights declare the chosen conversion.
    """
    if features.keys() != weights.keys():
        raise ValueError("each measured feature requires an explicit weight")
    if any(not isfinite(x) for x in list(features.values())
           + list(weights.values())):
        raise ValueError("features and weights must be finite")
    score = sum(weights[k] * features[k] for k in features)
    return _finite_result(score, "decision score")


def shadow_price_bounds(observations):
    """Bounds on lambda from (avoided_loss, extra_resource, accepted).

    Assumes lambda >= 0 and fixed score extra_resource - lambda * avoided_loss,
    with other effects held fixed. Bounds allow indifference at ties.
    No accepted/rejected choices give bounds 0/infinity respectively.
    """
    lower, upper = 0.0, float("inf")
    for avoided, resource, accepted in observations:
        if not (isfinite(avoided) and avoided > 0
                and isfinite(resource) and resource >= 0):
            raise ValueError("positive avoided loss and nonnegative cost required")
        if not isinstance(accepted, bool):
            raise ValueError("accepted must be a boolean")
        threshold = resource / avoided
        if not isfinite(threshold):
            raise ValueError("threshold exceeds finite arithmetic range")
        if accepted:
            lower = max(lower, threshold)
        else:
            upper = min(upper, threshold)
    return {"lower": lower, "upper": upper,
            "consistent": lower <= upper}


def policy_distance(first, second):
    """Total variation of two aligned finite action distributions.

    Absent action keys mean known zero probability. The caller supplies
    matched conditions; this function does not establish causal matching.
    Normalisation uses tolerance 1e-12; outputs have floating-point error.
    """
    for distribution in (first, second):
        if any(not isfinite(p) or p < 0 for p in distribution.values()):
            raise ValueError("probabilities must be finite and nonnegative")
```

```
        if not isclose(sum(distribution.values()), 1.0,
                       rel_tol=0.0, abs_tol=1e-12):
            raise ValueError("each distribution must sum to one")
    keys = first.keys() | second.keys()
    return 0.5 * sum(abs(first.get(k, 0) - second.get(k, 0)) for k in keys)


def valuation_gap(operative_candidates, model_admitted):
    """Finite-grid mismatch: operative valuation vs admitted self-description.

    Operative candidates satisfy the stated choice model. Admission status
    is known for every operative candidate; unknowns cannot be omissions.
    The rejection fraction is a uniform grid-count summary, not aversion.
    An empty operative set gives an undefined gap, not a measured mismatch.
    """
    operative = frozenset(operative_candidates)
    admitted = frozenset(model_admitted)
    excluded = operative - admitted
    if not operative:
        return {"excluded": excluded, "incompatible": None,
                "rejection_fraction": None}
    return {"excluded": excluded,
            "incompatible": not bool(operative & admitted),
            "rejection_fraction": len(excluded) / len(operative)}


def normal_response(mean, spread, shift=0.0, ambivalence_width=None):
    """Conditional fractions for a normally distributed decision margin.

    Positive margin favours the comparison action over the harmful action.
    A common nonnegative shift lowers the margin, holding other terms fixed.
    Width defines a near-zero-margin proxy, not psychological ambivalence.
    Parameters are declared, not fitted; outputs use floating-point CDFs.
    """
    if not (isfinite(mean) and isfinite(spread) and spread > 0
            and isfinite(shift) and shift >= 0):
        raise ValueError("finite mean, positive spread, nonnegative shift required")
    if ambivalence_width is not None and not (
            isfinite(ambivalence_width) and ambivalence_width > 0):
        raise ValueError("ambivalence width must be finite and positive")
    standard = NormalDist()

    def cdf_at(value):
        difference = value - mean
        if isfinite(difference):
            z = difference / spread
        else:
            # Opposite-sign finite operands can overflow subtraction.
            # Divide first in that case; an infinite z is a valid tail.
            z = value / spread - mean / spread
        return standard.cdf(z)

    near = None
    if ambivalence_width is not None:
        near = cdf_at(ambivalence_width) - cdf_at(-ambivalence_width)
    baseline = cdf_at(0.0)
    shifted = cdf_at(shift)
    return {"baseline_fraction": baseline,
            "shifted_fraction": shifted,
            "newly_selected_fraction": shifted - baseline,
            "near_zero_fraction": near}
```

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